

Robust Target-Specific Recoverability Maps for Reduced-Lead ECG Reconstruction

Evidence-gated extended protocol and legacy research record

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Abstract

Reduced-lead electrocardiogram (ECG) reconstruction is commonly assessed only after a model produces a waveform. We propose a training-set diagnostic that maps each observed configuration, waveform segment, and target lead to a continuous recoverability score. The score combines target geometry outside the observed-row span with within-span amplification, then aggregates over a prespecified family of rank, lead-weighting, and patient-bootstrap fits. The sole primary empirical claim is that this robust map predicts held-out target-lead reconstruction error beyond simple configuration and signal-scale covariates, across reconstruction families and on an external cohort. Numerical claims are withheld pending the Stage 15 evidence decision. This extended document separates the new protocol from earlier conditional theory, calibration, clinical-event, lead-selection, and diffusion branches, which are retained only as a transparent legacy supplement.

1 Research question and contribution

Reduced-lead acquisition motivates fixed transforms, regression, and learned completion models [1–8]. These methods answer how to synthesize a missing waveform. We ask a complementary question: before selecting a reconstructor, can training-set spatial geometry anticipate which segment and target lead will have larger held-out error?

The refactored paper has one claim and one falsification route. For each observed set S , segment s , and target lead ℓ , a frozen robust score should (i) rank held-out target error, (ii) add out-of-sample predictive value beyond lead count, configuration rank, target scale, pairwise correlation, and global conditioning, (iii) retain its direction across transparent and learned reconstructors, and (iv) retain its direction on a second cohort. Failure of the incremental, cross-model, or external tests prevents a positive headline. The score is described as a map or diagnostic; it is not a clinical guarantee or a claim about all information in an ECG waveform.

2 Robust target-specific map

Let $\mathbf{x}_{i,s}(t) \in \mathbb{R}^p$ be the multilead signal for patient i in segment s . A training-only weighted covariance fit gives $\mathbf{M}_s^v \in \mathbb{R}^{p \times r}$, with $v = (r, w, b)$ indexing candidate rank, lead weighting, and

patient-bootstrap replicate. For target row $\mathbf{m}_{s,\ell}^v$ and observed rows $\mathbf{M}_{s,S}^v$, define

$$u_{s,\ell}^v(S) = \frac{\|\mathbf{m}_{s,\ell}^v[\mathbf{I} - (\mathbf{M}_{s,S}^v)^+ \mathbf{M}_{s,S}^v]\|_2}{\|\mathbf{m}_{s,\ell}^v\|_2 + \epsilon}, \quad (1)$$

$$g_{s,\ell}^v(S) = \|\mathbf{m}_{s,\ell}^v(\mathbf{M}_{s,S}^v)^+\|_2, \quad (2)$$

$$q_{s,\ell}^v(S) = \left[(u_{s,\ell}^v)^2 + \left\{ \frac{\nu_{s,S}^v g_{s,\ell}^v}{\sigma_{s,\ell}^v + \epsilon} \right\}^2 \right]^{1/2}, \quad (3)$$

where ν is a training-only observed residual scale and σ a training-only target scale. The first term measures target geometry absent from the observed-row span of the fitted representation; the second measures amplification relative to target scale. The primary map is the median q^v over plausible fits, accompanied by patient-bootstrap quantiles. Its empirical value comes only from the held-out tests below.

Prespecified perturbations. The primary fit uses eight algebraically independent measured channels; the 12 displayed leads form a weighting sensitivity. Folds 1–7 fit rank candidates 2–5, fold 8 chooses the primary rank by an out-of-sample representation criterion, and all plausible ranks remain in the envelope. Patient-cluster bootstrap resamples refit the subspace. Sign conventions, units, segment windows, pseudoinverse tolerance, and missing-segment rules are frozen before fold 10 is opened.

3 Evaluation and analysis contract

Split discipline and cohorts. PTB-XL official folds 1–7 are used for fitting, fold 8 for design choices, fold 9 for a full pipeline rehearsal without claim selection, and fold 10 once for the primary evaluation [9]. The Chapman–Shaoxing–Ningbo cohort is the external test and uses the frozen PTB-XL specification [10]. The patient is the resampling unit. QRS, ST, and T form the core analysis; P-wave results move to the supplement if the frozen delineation-quality criterion is not met.

Reconstruction families. Prior mean and ridge regression are transparent anchors. A mask-conditioned 1-D U-Net represents a flexible learned family. Stage 15 also requires at least one reproducible public strong-method adapter; ECGrecover and ImputeECG are prioritized subject to artifact, license, and protocol compatibility [6, 7]. All reconstructors receive the same observed samples and mask. Seeds, checkpoints, preprocessing, failures, and unavailable-method reasons are retained.

Primary endpoint and test. The primary outcome is patient-level target-lead normalized RMSE within segment. For $(S, s, \ell, \text{reconstructor})$ cells, we first report rank association with the frozen map. We then compare a baseline predictive model containing lead count, configuration rank, target RMS, maximum pairwise correlation, and global conditioning against the same model plus q . Incremental out-of-sample fit, coefficient direction, and patient-cluster bootstrap intervals are co-primary evidence. Segment- and model-specific contrasts are secondary and multiplicity controlled. External validation tests the frozen score–error relationship; subspace-angle agreement alone is insufficient.

Table 1: Primary claim–evidence matrix before the Stage 15 decision.

Evidence item	Gate value	Failure consequence
Held-out score–error association	PENDING–STAGE 15	No positive primary claim
Increment beyond simple covariates	PENDING–STAGE 15	Recast as descriptive map
Rank/weight/bootstrap stability	PENDING–STAGE 15	Narrow supported scope
External-cohort association	PENDING–STAGE 15	No transfer language
Coverage across model families	PENDING–STAGE 15	Model-specific conclusion
Patient-level uncertainty	PENDING–STAGE 15	Evidence gate fails

4 Evidence gates, reproducibility, and decision policy

Stage 5 requires a verified literature ledger and a novelty statement that distinguishes the map from reconstruction methods. Stage 9 freezes splits, endpoints, covariates, reconstructors, perturbations, exclusions, seeds, and decision rules. Stage 15 permits PROCEED only after a real non-simulated run provides immutable run identifiers, clean source state, dataset and split hashes, checkpoint hashes, leakage tests, patient-cluster uncertainty, a strong public baseline, external validation, and a completed claim–evidence matrix. Stage 20 additionally requires no pending macros, verified citations, readable figures, a clean manuscript build, and a strict release audit. REFINER and PIVOT are valid outcomes; negative and failed runs remain in the record. The detailed contract is maintained in `docs/research_protocol.md`.

5 Limitations and current status

The map is built from a fitted linear representation, while flexible models may exploit nonlinear population regularities. Delineation, pathology mix, devices, noise, and lead placement can change both geometry and error. One external cohort is not deployment validation, and waveform similarity is not diagnostic or clinical equivalence. The current empirical status is deliberately PENDING–STAGE 15: the protocol is ready for gated execution, but it does not yet support a numerical abstract or a positive performance conclusion.

6 Conclusion

The intended contribution is a narrow, falsifiable diagnostic: a robust target-specific map that predicts held-out reconstruction difficulty beyond simple baselines and complements direct model benchmarking. Its headline is conditional on the Stage 15 evidence decision. The following pages retain earlier branches for auditability, separated from the primary narrative and evidence chain.

A Legacy supplement: conditional fixed-rank analysis

Legacy supplement—not evidence for the primary claim. This material records an earlier research branch. It is excluded from the Stage 15 claim–evidence decision and from the four-page submission.

The material in this section assumes a fixed low-rank representation and studies conditional algebraic properties. It is retained to preserve research history, not to justify the robust map–error hypothesis in the main paper.

Model. Within segment s write the population 12-lead vector as $\mathbf{L}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{d} + \mathbf{r}_s$, where $\mathbf{M}_s \in \mathbb{R}^{12 \times 3}$ is the (population-estimated, PCA) dipolar basis, $\mathbf{d}$ the dipole coordinates, and $\mathbf{r}_s$ the non-dipolar residual. Observing S gives $\mathbf{y}_S = \mathbf{Sel}_S \mathbf{L}_s + \mathbf{n}$, with $\mathbf{M}_{s,S} = \mathbf{Sel}_S \mathbf{M}_s$. Write the exact Moore–Penrose pseudo-inverse $\mathbf{M}_{s,S}^+$ and the exact row-space projector $\mathbf{P}_{\text{obs}} = \mathbf{M}_{s,S}^+ \mathbf{M}_{s,S}$. For numerical deployment we also use the *rank- r truncation* at relative tolerance ϱ , $\mathbf{M}_{s,S}^{(\varrho)} = \mathbf{U}_r \boldsymbol{\Sigma}_r \mathbf{V}_r^\top$ keeping singular triplets with $\sigma_i > \varrho \sigma_{\max}$, with pseudo-inverse $\mathbf{M}_{s,S}^{(\varrho)+}$ and projector $\mathbf{P}_{\text{obs}}^{(\varrho)} = \mathbf{V}_r \mathbf{V}_r^\top$.

Proposition 1 (Exact per-target-lead dipolar identifiability and conditioning). *For each target lead ℓ define $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})\|_2$ and $\kappa_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$ from the exact $\mathbf{M}_{s,S}$.*

- (i) *Identifiability.* $\eta_{s,\ell}(S) = 0$ iff $\mathbf{e}_\ell^\top \mathbf{M}_s$ lies in the exact row space of $\mathbf{M}_{s,S}$; then $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$ is a deterministic function of $\mathbf{M}_{s,S} \mathbf{d}$ —two dipoles with the same observation have the same lead- ℓ dipolar value, at any SNR. If $\eta_{s,\ell}(S) > 0$ there is a $\boldsymbol{\delta} \in \ker \mathbf{M}_{s,S}$ (an unobserved dipole direction) with $\mathbf{e}_\ell^\top \mathbf{M}_s \boldsymbol{\delta} \neq 0$; then $\mathbf{d}$ and $\mathbf{d} + t\boldsymbol{\delta}$ produce identical observations for all t while lead ℓ ’s dipolar value changes, so it is not identifiable at any SNR.
- (ii) *Conditioning.* On the identifiable part, the lead- ℓ error from observation noise and observed non-dipolar residual obeys $|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+ (\mathbf{Sel}_S \mathbf{r}_s + \mathbf{n})| \leq \kappa_{s,\ell}(S) (\|\mathbf{Sel}_S \mathbf{r}_s\|_2 + \|\mathbf{n}\|_2)$.

The global $\kappa_s(S) = \|\mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$ (spectral norm) upper-bounds every per-lead $\kappa_{s,\ell}(S)$ (a row norm never exceeds the spectral norm; it is not their maximum).

Exact vs. ϱ -identifiability (numerical deployment). The deployed map computes $\eta_{s,\ell}^{(\varrho)}, \kappa_{s,\ell}^{(\varrho)}$ from the truncation $\mathbf{P}_{\text{obs}}^{(\varrho)}$, so a singular direction observed only through a value below $\varrho \sigma_{\max}$ is treated as unobserved. Then $\eta_{s,\ell}^{(\varrho)} > 0$ means lead ℓ is *not stably recoverable at resolution ϱ* (outside the retained numerical row space); it does *not* by itself imply the exact matrix has a null direction—a tiny-but-nonzero singular value is exactly identifiable ($\eta = 0$) yet ϱ -unidentifiable ($\eta^{(\varrho)} > 0$), as we verify numerically (`tests/test_exact_vs_rho.py`). We therefore call the $\eta^{(\varrho)}$ heatmap the *ϱ -identifiability* map; for well-separated configurations the exact and ϱ verdicts coincide, and near-rank-deficient S are reported with a ϱ -sweep and record-bootstrap CIs. **The limb-6 flagship is exact:** the six limb leads are exact linear combinations of I and II (Einthoven/Goldberger), so $\mathbf{M}_{s,S}$ has exact rank 2 and precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of ϱ . This exactness is conditional on the a-priori rank-3 dipole model for $\mathbf{M}_s$: at retained rank 2 the limb-6 observation spans all of $\mathbf{M}_s$ and the precordial verdict collapses to $\eta=0$; the non-identifiability appears at rank ≥ 3 (the third dipolar direction), so “exact” means exact *given* rank 3.

Relation to classical estimability. Part (i) is the classical estimability criterion (Gauss–Markov / [11]) for the linear functional $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$; we claim no novelty for this identity. The contribution (Sec. C) is its use as a graded, empirically-estimated recoverability map with per-record calibrated error rates (Sec. B).

The non-dipolar residual (no independence claim on real data). Let $\mathbf{m}_s(\mathbf{y}_S) = \mathbb{E}[\mathbf{r}_s \mid \mathbf{y}_S]$ and $\mathbf{u}_s = \mathbf{r}_s - \mathbf{m}_s(\mathbf{y}_S)$, so $\mathbb{E}[\mathbf{u}_s \mid \mathbf{y}_S] = \mathbf{0}$. We do *not* assume $\mathbf{u}_s \perp \mathbf{y}_S$. Under squared loss the Bayes-optimal reconstruction error of the (vector) residual is

$$\inf_{\hat{\mathbf{r}}} \mathbb{E} \|\mathbf{r}_s - \hat{\mathbf{r}}(\mathbf{y}_S)\|_2^2 = \mathbb{E}[\text{tr Cov}(\mathbf{r}_s \mid \mathbf{y}_S)].$$

Part of $\mathbf{r}_s$ (the *empirically predictable residual*) is recovered by a predictor trained on $\mathbf{y}_S$ and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—*not* declared unrecoverable a priori.

Proposition 2 (Strict independence limit – synthetic model only). *If the data are generated so that a component $\mathbf{u}$ is statistically independent of $\mathbf{y}_S$ (an explicit synthetic construction), then $p(\mathbf{u} \mid \mathbf{y}_S) = p(\mathbf{u})$, the Bayes estimate of $\mathbf{u}$ under squared loss is its prior mean, and $\mathbb{E}\|\hat{\mathbf{u}} - \mathbf{u}\|_2^2 \geq \text{tr Cov}(\mathbf{u})$ for any estimator. This strict limit is invoked only for the synthetic model; on real ECG we make no such independence claim.*

An exact minimax recoverability floor

Proposition 1 says *whether* a target lead’s dipolar coordinate is recoverable ($\eta=0$) and *how well-conditioned* the estimate is (κ). We now give the *exact* minimax risk of estimating the target’s dipolar functional from the exact (noise-free) observation, over a second-moment class of dipole priors—turning the binary verdict into a per-lead risk in physical units. Fix segment s and observed set S ; let $\mathbf{d} \in \mathbb{R}^3$ be the dipole coordinate, $\mathbf{y}_S = \mathbf{M}_{s,S}\mathbf{d}$ the exact observation, and $\theta_\ell = \mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$ the target dipolar functional of lead ℓ . Write the (orthogonal, symmetric) observed row-space projector $\mathbf{P} = \mathbf{M}_{s,S}^+ \mathbf{M}_{s,S}$, $\mathbf{Q} = \mathbf{I} - \mathbf{P}$, and

$$a_{s,\ell}(S)^2 = \mathbf{e}_\ell^\top \mathbf{M}_s \Sigma_{Q|P} \mathbf{M}_s^\top \mathbf{e}_\ell, \quad \Sigma_{Q|P} = \mathbf{Q} \Sigma_d \mathbf{Q} - \mathbf{Q} \Sigma_d \mathbf{P} (\mathbf{P} \Sigma_d \mathbf{P})^+ \mathbf{P} \Sigma_d \mathbf{Q},$$

where $\Sigma_d \succ \mathbf{0}$ is a fixed dipole second-moment bound.

Proposition 3 (Exact minimax dipolar floor). *Let $\mathcal{P}(\Sigma_d) = \{\pi : \mathbb{E}_\pi[\mathbf{d}] = \mathbf{0}, \mathbb{E}_\pi[\mathbf{d}\mathbf{d}^\top] \preceq \Sigma_d\}$. For the exact observation $\mathbf{y}_S = \mathbf{M}_{s,S}\mathbf{d}$, the minimax risk over all measurable (possibly nonlinear) estimators $\hat{\theta}_\ell(\mathbf{y}_S)$ of θ_ℓ satisfies*

$$\inf_{\hat{\theta}_\ell} \sup_{\pi \in \mathcal{P}(\Sigma_d)} \mathbb{E}_\pi[(\hat{\theta}_\ell(\mathbf{y}_S) - \theta_\ell)^2] = a_{s,\ell}(S)^2,$$

attained by the Gaussian least-favourable prior $\pi^ = \mathcal{N}(\mathbf{0}, \Sigma_d)$ and a linear estimator. Consequently $a_{s,\ell}(S) = 0$ iff $\eta_{s,\ell}(S) = 0$.*

Proof. Upper bound. For a coefficient $\mathbf{c}$ the linear estimator $\mathbf{c}^\top \mathbf{y}_S$ has error $(\mathbf{c}^\top \mathbf{M}_{s,S} - \mathbf{e}_\ell^\top \mathbf{M}_s) \mathbf{d} = \mathbf{g}_\mathbf{c}^\top \mathbf{d}$, so for every $\pi \in \mathcal{P}(\Sigma_d)$, $\mathbb{E}_\pi[(\mathbf{c}^\top \mathbf{y}_S - \theta_\ell)^2] = \mathbf{g}_\mathbf{c}^\top \mathbb{E}_\pi[\mathbf{d}\mathbf{d}^\top] \mathbf{g}_\mathbf{c} \leq \mathbf{g}_\mathbf{c}^\top \Sigma_d \mathbf{g}_\mathbf{c}$, with equality at π^* . As $\mathbf{c}$ varies, $\mathbf{M}_{s,S}^\top \mathbf{c}$ ranges over range $\mathbf{P}$, so $\min_{\mathbf{c}} \mathbf{g}_\mathbf{c}^\top \Sigma_d \mathbf{g}_\mathbf{c}$ is the Σ_d -projection residual of $\mathbf{w} := \mathbf{M}_s^\top \mathbf{e}_\ell$ onto range $\mathbf{P}$, $= \mathbf{w}^\top [\Sigma_d - \Sigma_d \mathbf{P} (\mathbf{P} \Sigma_d \mathbf{P})^+ \mathbf{P} \Sigma_d] \mathbf{w}$. Since conditioning on $\mathbf{P}\mathbf{d}$ removes all $\mathbf{P}$ -variance, $\Sigma_d - \Sigma_d \mathbf{P} (\mathbf{P} \Sigma_d \mathbf{P})^+ \mathbf{P} \Sigma_d = \Sigma_{Q|P}$, so this equals $a_{s,\ell}(S)^2$. Hence a linear estimator has worst-case risk $a_{s,\ell}(S)^2$ and the minimax risk is $\leq a_{s,\ell}(S)^2$. **Lower bound.** The minimax risk is at least the Bayes risk under any single prior in the class; take π^* . Its Bayes risk—the infimum over *all* estimators of $\mathbb{E}_{\pi^*}[(\hat{\theta}_\ell - \theta_\ell)^2]$ —is attained by the posterior mean and equals $\text{Var}_{\pi^*}(\theta_\ell \mid \mathbf{y}_S) = a_{s,\ell}(S)^2$ (Gaussian conditioning: $\mathbf{y}_S$ fixes $\mathbf{P}\mathbf{d}$ and the residual variance of $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{Q}\mathbf{d}$ is the Schur complement

$\Sigma_{Q|P}$). Thus the minimax risk is $\geq a_{s,\ell}(S)^2$, and the two bounds coincide. $a=0 \iff \eta=0$. $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{Q}\|_2$. If $\eta=0$ then $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{Q} = \mathbf{0}$, so $a^2 = 0$. If $\eta>0$ then $\mathbf{v} := \mathbf{Q}^\top \mathbf{M}_s^\top \mathbf{e}_\ell \neq \mathbf{0}$; because $\Sigma_d \succ \mathbf{0}$ the Schur complement $\Sigma_{Q|P}$ is positive definite on $\text{range } \mathbf{Q}$, so $a^2 = \mathbf{v}^\top \Sigma_{Q|P} \mathbf{v} > 0$. $\square$

Scope, and what the tests check. The result is *exact* and *noise-free*: the minimax risk of the dipolar functional θ_ℓ over the second-moment class $\mathcal{P}(\Sigma_d)$ under an exact observation, with the infimum over all measurable estimators. We deliberately do *not* state a noise-inflated or finite-sample “floor.” The observed-noise amplification $\kappa_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$ (Prop. 1) is an *upper* bound on how observed noise/residual propagates into the estimate, not a lower risk floor; and a bootstrap minimum of $a_{s,\ell}$ over the estimated $\widehat{\mathbf{M}}_s$ is not a certified uniform confidence bound—so neither is claimed as part of the theorem. The certified object is the *dipolar* functional; a floor on the full lead would require bounding the non-dipolar residual, which we instead meter with the distribution-free intervals of Sec. B. `tests/test_floor.py` numerically checks that *selected* estimators (the posterior mean, ridge, OLS) do not beat $a_{s,\ell}$ under π^* : this validates the theorem on those estimators, it does not re-prove it.

Operational value: $a_{s,\ell}$ predicts error. The floor is a *worst-case-prior* quantity, not an empirical envelope. On real ECG its value is as a *predictor*: $a_{s,\ell}$ strongly predicts and ranks the measured per-lead dipolar error (Spearman 0.81, Pearson 0.93; Fig. 1) and the identifiable/unidentifiable split is separated with AUC 0.87. Because it is worst-case over $\mathcal{P}(\Sigma_d)$, estimators exploiting the true (non-Gaussian) population structure fall *below* it on a fraction of the 84 non-trivial ($\eta>0$) cells (the pure-dipolar reconstructor on 0.21, ridge/OLS on 0.60). This below-floor gap and the Tier-II residual used by the calibration analysis in Sec. B are *different objects*: the former compares an estimator’s empirical-population risk with a worst-case-prior benchmark, whereas the latter is a record-level prediction error under a specified reconstructor and feature definition. No equality, decomposition, or coverage relationship follows between them. Any empirical association would require a separately frozen held-out analysis. The observed below-floor cases remain consistent with the theorem, which concerns the worst case rather than the risk under a single empirical population. The η -split likewise predicts a trained diffusion model’s error (median 0.066 vs 0.227 mV for $\eta=0$ vs $\eta>0$; `experiments/certificate_floor_diffusion.py`).

Cross-cohort sensitivity of the verdict

The certificate is computed from a *fitted* dipolar subspace $\mathbf{M}_s$, so it is natural to ask how the verdict moves across cohorts. One fact is exact and useful: $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})\|_2$ depends on $\mathbf{M}_s$ *only through its column space*. Replacing $\mathbf{M}_s$ by $\mathbf{M}_s \mathbf{R}$ for any $\mathbf{R} \in O(3)$ rotates $\mathbf{P}_{\text{obs}}$ and $\mathbf{e}_\ell^\top \mathbf{M}_s$ together and leaves every η unchanged; hence two cohorts that share a dipolar subspace (all principal angles zero) have identical verdicts, and in general the drift is governed by how far the two subspaces separate.

A $\sin \Theta$ (Davis–Kahan [12]) perturbation makes this quantitative but, at the conditioning we observe, *vacuously so*. After optimal alignment, $|\eta_{s,\ell}^A(S) - \eta_{s,\ell}^B(S)| \leq 2 \sin(\theta^*/2) (1 + 2 \kappa_s)$, where θ^* is the largest principal angle between the two cohorts’ segment- s dipolar subspaces and $\kappa_s = \max(\kappa_s^A, \kappa_s^B)$ is the larger observed-set conditioning; the projector-difference step uses the Stewart–Wedin bound and assumes the observed block keeps its rank across cohorts and that the ϱ -truncated projector coincides with the exact one. On PTB-XL→Chapman (limb-6→V2) the right-hand side evaluates to 1.37 (QRS) and 32.2 (ST), *both exceeding* the trivial bound $|\Delta\eta| \leq 1$ (since $\eta \in [0, 1]$). The inequality therefore **certifies nothing here**: it does not establish that the QRS verdict transfers, and it is not a pre-deployment guarantee. A non-vacuous statement would need a target-

specific bound built from the aligned target-lead rows and the exact projector difference; we leave that to future work and do not claim it.

We accordingly present the cross-cohort comparison as *empirical sensitivity*, not a certified bound (`results/transfer_bound.json`). The measured principal angle and the measured verdict drift agree qualitatively: for QRS the two dipolar subspaces are nearly aligned ($\theta^*=13.8^\circ$; η drift 0.033), whereas for ST they are nearly orthogonal ($\theta^*=85.7^\circ$, CI up to 89.6°) with a larger but highly uncertain drift (the Chapman η bootstrap CI is $[0.02, 0.71]$). This is a two-cohort observation—the QRS dipolar geometry happens to be shared between these cohorts while the low-amplitude ST geometry is not, consistent with the subspace-angle ablations of Sec. D—not a guarantee about unseen cohorts.

B Legacy supplement: calibration branch

Legacy supplement—not evidence for the primary claim. This material records an earlier research branch. It is excluded from the Stage 15 claim–evidence decision and from the four-page submission.

The earlier conformal-calibration branch addresses a different estimand from the robust score and is outside the primary claim.

B.1 Calibrated intervals for the predictable residual

The prediction target is the *record-level segment-mean* off-dipole residual of the target lead (one scalar per record: the mean over the segment’s samples). We train a gradient-boosted quantile regressor of it on the observed leads’ segment means and apply Mondrian conformalized quantile regression [13] per (S, s, ℓ) group, with strict fold discipline: basis and model on folds 1–7, the regressor’s `max_iter` selected by pinball loss on fold 8, conformal calibration on fold 9, and a single evaluation on fold 10 (nominal $1-\alpha=0.90$). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered (S, s, ℓ) groups, fold-10 within-group coverage clusters near nominal (0.87–0.94; e.g. {I,II,V1,V3,V5} QRS/V2 0.91, record-bootstrap CI $[0.89, 0.93]$, mean width 0.150 mV).

What is and is not guaranteed. Mondrian CQR gives *within- (S, s, ℓ) -group marginal* coverage under exchangeability—not coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.72 ($n=85$), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

C Legacy supplement: PTB-XL map, event, and baseline studies

Legacy supplement—not evidence for the primary claim. This material records an earlier research branch. It is excluded from the Stage 15 claim–evidence decision and from the four-page submission.

These analyses use the earlier fixed-rank formulation. Their values are historical artifacts and must not populate the refactored manuscript without a new Stage 9 freeze and Stage 15 decision.

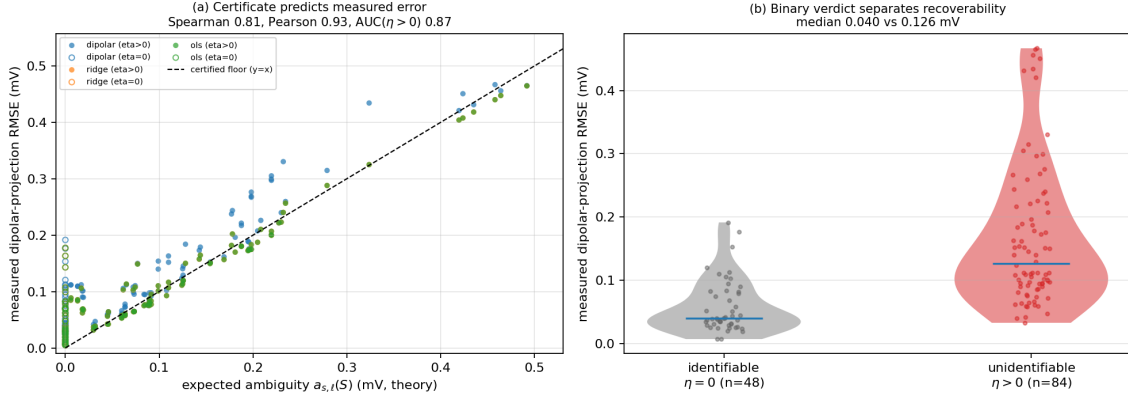


Figure 1: Legacy diagnostic retained for traceability. It is not evidence for the robust map–error claim and is excluded from the short submission.

C.1 The recoverability map on PTB-XL

We estimate $\mathbf{M}_s$ (rank 3; the cardiac dipole is 3-D *a priori*, and the cumulative explained variance confirms rank 3 captures the bulk—a rank-sensitivity sweep $r=2\dots 5$ is reported in `results/recoverability_maps.json`) per segment on the training NORM records, with RECORD-LEVEL bootstrap CIs (resample records, refit $\mathbf{M}_s$; $\varrho=10^{-2}$; Fig. 2). We plot the *normalized* identifiability $\tilde{\eta}_{s,\ell} = \eta_{s,\ell} / \|\mathbf{e}_\ell^\top \mathbf{M}_s\|_2$ —a dimensionless relative unobserved geometric gain (a ratio of L_2 norms, not a variance fraction). A single observed lead leaves every other lead unidentifiable; a dipole-spanning $\{\text{I,II,V2}\}$ or $\{\text{I,II,V1,V3,V5}\}$ makes all targets *exactly* identifiable ($\eta=0$).

The robust flagship is exact. The six limb leads are exact linear combinations of I and II, so $\mathbf{M}_{s,S}$ for limb-6 has *exact* rank 2: precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of ϱ (Prop. 1).

Graded ordering: stable for QRS, weighting-sensitive for ST/T. A duplicated-limb sensitivity analysis (`results/lead_weighting.json`: refit on the 8 algebraically independent leads, on the *same* records) shows the V1–V6 $\tilde{\eta}$ ordering is bootstrap-stable for QRS (Spearman 1.00, CI lower bound 0.88; anterior-before-lateral in 0.98 of resamples) but *not* robust for the low-amplitude ST/T segments (ST Spearman point 0.89 with a wide bootstrap CI; anterior-before-lateral in only 0.77 of resamples). We therefore report the ST/T $\tilde{\eta}$ /ambiguity magnitudes with this caveat and *do not* claim a fine ST/T ordering; the binary limb→precordial verdict above is the robust ST statement. The estimated subspace also depends on diagnosis (a NORM vs. MI principal-angle sensitivity is reported), so the map is a *NORM-trained reference*.

Downstream ST cost across reconstructors. We reconstruct the full 10-second waveform with three real continuous linear reconstructors (observed leads exact; ST at J+60 ms vs. the isoelectric PR-segment baseline [P offset – QRS onset], on fiducials from the observed Lead II shared between truth and reconstruction; median over beats; the PR-segment baseline fell back to the pre-QRS window in $\approx 14\%$ of beats), and count ST-threshold events ($|\text{ST}| \geq 0.1 \text{ mV}$; false positive = crossing in reconstruction not truth, false negative = vice versa; Table 2). On 1498 common records the *total* wrong-event rate has point estimates ranging 51.1–53.7% across reconstructors and is *dominated by false negatives* (missed true crossings: limb→precordial recovery smooths ST toward the population), with the false-positive/false-negative split a reconstructor choice (paired record-bootstrap Δ on the FP/FN split, and on the total, in `results/st_safety.json`). We do not derive a minimax/Bayes bound and make no equivalence claim: we report the range of total-error

Target-specific recoverability map (grey hatched = observed lead)

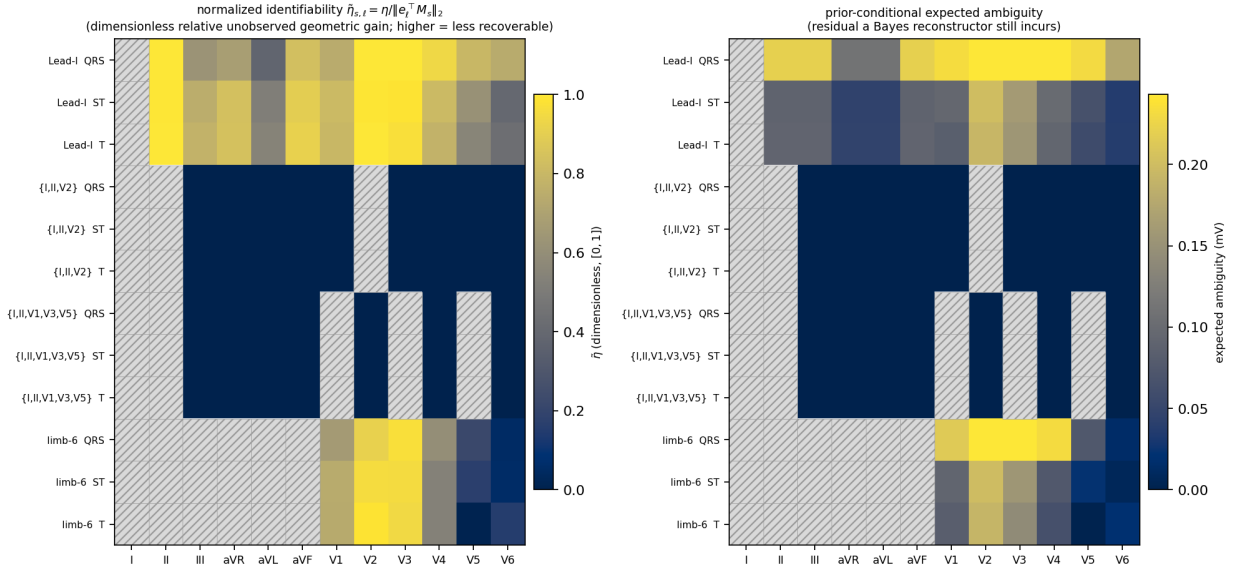


Figure 2: Recoverability map: normalized identifiability $\tilde{\eta}_{s,\ell}(S)$ (left, dimensionless) and prior-conditional expected ambiguity in mV (right), colorblind-safe `cividis`; grey hatched = observed. For limb-6 all precordial leads are unidentifiable ($\eta > 0$); the fine ST/T ordering is weighting-sensitive (text).

Table 2: Limb-6 $\rightarrow$ precordial (all V1–V6) ST reconstruction, 1498 common fold-10 records. FP/FN = false-positive/false-negative ST-threshold crossings (%); total = FP+FN. FN dominates (missed crossings); the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

reconstructor	FP %	FN %	total %	ST err (mV)
dipolar	5.3	45.7	51.1	0.077
ridge	1.7	52.1	53.7	0.076
OLS	1.7	51.8	53.5	0.076

point estimates, *not* a certified lower bound, and ST-threshold events, not diagnoses. Because the PR-segment baseline fell back to the pre-QRS window in $\approx 14\%$ of beats, the exact ST magnitudes inherit that delineation limitation.

Truncation-tolerance sensitivity. Well-posed configurations are ϱ -stable ($\{I, II, V1, V3, V5\}$ is rank 3 across $\varrho \in \{10^{-4}, \dots, 10^{-1}\}$); near-rank -deficient sets are ϱ -sensitive and reported with the ϱ -sweep and bootstrap CIs. A tiny-but-nonzero singular value is exactly identifiable yet ϱ -unidentifiable, which is why we separate exact from ϱ -identifiability (Prop. 1).

C.2 Baselines and what the map predicts

Table 3 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: the same shared split (train folds 1–7, ridge λ selected on fold 8, test on NORM fold-10), the same 800 test records, per-timepoint waveform RMSE on the same segment indices, and *paired* record-bootstrap CIs on the step-to-step differences. We include a *representative* masked-completion baseline—an arbitrary-mask conditional 1-D U-Net trained with random lead

Table 3: Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol on NORM fold-10 (800 test records, shared split). “dipolar” = Tier I; “ridge” = per-segment ridge (Tier I+II); “U-Net” = representative masked-completion baseline (3-seed mean). Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
{I,II,V2}	0.467	0.236	0.220	0.183
{I,II,V1,V3,V5}	0.481	0.183	0.167	0.164
limb-6	0.481	0.361	0.344	0.238

masks (one model serves any configuration, no target-lead leakage), the same family as [4, 6, 7], run over three seeds (mean reported)—not a state-of-the-art claim.

On a dipole-spanning set the methods form a ladder in point estimates (the ridge→U-Net step is within its paired CI)—prior-mean → dipolar (Tier I) → per-segment ridge (Tier I+II) → U-Net—evidence of recoverable non-dipolar content (the calibrated Tier II layer), though the neural model’s margin over a simple ridge is small (QRS 0.164 vs. 0.167 mV on {I,II,V1,V3,V5}; paired Δ CI $[-0.007, +0.000]$ mV). On **limb-6** the U-Net still *lowers* error over the prior mean (QRS 0.481 → 0.238)—it recovers the predictable, largely population-correlated structure—but plateaus at $1.46\times$ its spanning-set error (paired spanning→limb-6 Δ CI $[+0.070, +0.079]$ mV) and no reconstructor closes that gap. This residual gap is *consistent with* the unobserved dipolar coordinate the map’s $\eta > 0$ marks as unidentifiable (we do not decompose the error into projected components, so we claim consistency, not equality): capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing it. We therefore scope the identifiability claim to the dipolar component, not to a blanket “not recoverable.”

Empirical, not physical, subspace. On real PTB-XL the estimated $\mathbf{M}_s$ has three graded principal angles to the classical inverse-Dower vectorcardiographic space [1, 2]: one close ($2\text{--}6^\circ$), one moderate ($10\text{--}15^\circ$), and one substantially different ($43\text{--}55^\circ$, bootstrap CIs excluding small angles). We therefore treat $\mathbf{M}_s$ as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all recoverability quantities are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment (and robust to the duplicated-limb weighting).

D Legacy supplement: sensitivity and cross-cohort branches

Legacy supplement—not evidence for the primary claim. This material records an earlier research branch. It is excluded from the Stage 15 claim–evidence decision and from the four-page submission.

Earlier subspace-angle and cross-cohort summaries are exploratory sensitivity analyses. The new protocol instead requires external validation of the score–error relationship.

E Cross-Dataset Transfer (Matched Comparison)

To test whether the certificate is an artifact of one cohort we re-estimate $\mathbf{M}_s$ under the same unified truncated-SVD numerics on the independent Chapman-Shaoxing-Ningbo 12-lead database [10] and compare principal angles, rank, and κ to PTB-XL. To avoid confounding *cohort* with *case mix*, we run a **matched** comparison (PTB-XL NORM vs. Chapman sinus-rhythm records) alongside an all-record comparison. Chapman records are sampled *randomly* from the complete manifest with a fixed seed (not a deterministic folder/name stride), channels are reordered by WFDB `sig_name` with a 12-lead assertion and resampled from the record’s actual sampling rate, and the exact loaded manifest is hashed (`results/cross_dataset.json`); we used 350 Chapman records, of which 81 were sinus-rhythm.

QRS transfers; ST/T do not. In the *matched* (normal-vs-normal) comparison the QRS subspaces are close (max principal angle 14° , record-bootstrap CI [10, 25]), and the well-posed conditioning transfers (κ for {I,II,V1,V3,V5}: 2.4 PTB-XL vs. 2.3 Chapman). limb-6 is effective rank 2 on both cohorts (we do not invert a near-zero singular value into a spurious 10^5 -scale κ). The low-amplitude segments differ sharply even under the matched comparison (ST max angle 86° , T 81°), and the all-record comparison is similar (QRS 14° , ST 81°). Because the ST/T difference *survives the matched case mix*, we describe it as cohort-specific: the QRS map and the limb-6 verdict are cohort-stable, while the ST/T third spatial direction is cohort-specific and should be re-estimated per dataset.

Duplicated-limb weighting (`results/lead_weighting.json`). On the *same* records as the primary map, refitting on the 8 algebraically-independent leads leaves the identifiability *verdict* unchanged (spanning-set η identical; limb-6 precordial stays unidentifiable) and the QRS $\tilde{\eta}$ ordering bootstrap-stable, but the low-amplitude ST/T ordering is *not* robust to the weighting (Sec. C); we therefore make no fine ST/T ordering claim under either fit.

F Legacy supplement: active lead-selection branch

Legacy supplement—not evidence for the primary claim. This material records an earlier research branch. It is excluded from the Stage 15 claim–evidence decision and from the four-page submission.

The earlier greedy lead-selection analysis optimized a score fitted on one cohort and did not model electrode placement constraints. It is not part of the current contribution and makes no selection or approximation claim. A future prescriptive study would require a separate Stage 9 protocol, bootstrap and cross-cohort stability, and physical acquisition constraints.

G Legacy supplement: diffusion and null-space branch

Legacy supplement—not evidence for the primary claim. This material records an earlier research branch. It is excluded from the Stage 15 claim–evidence decision and from the four-page submission.

The following negative-result record concerns a different generative-model question. It is retained for transparency and cannot be used to support the primary robust-map claim.

H A Negative Result: the Generative “Fabrication-Objective” Story Did Not Survive

This project began with a stronger claim—that a diffusion reconstructor would demonstrate “fabrication is a property of the objective.” A pre-submission re-analysis abandoned it. We document the abandonment in full because the negative result is itself informative and because selective reporting of the earlier (favourable) numbers would be exactly the failure the certificate is meant to prevent.

Setup. An arbitrary-mask conditional DDPM (12-channel binary mask + observed values; random-mask training so one model serves any configuration, with explicit leakage tests) reconstructs $\{V2, V4, V6\}$ from $\{I, II, V1, V3, V5\}$ under classifier-free-style guidance weight $w \in \{1, 2, 4, 6\}$ (results/gpu_deficit_ci.json, realism_metrics.json, results/gpu_oracle_gate.json; all lineage-stamped, numbers auto-generated).

(1) The original effect was a configuration-leakage artifact. The earlier exhibit reused one model trained on a fixed observed set to score a *different* configuration. With leakage removed (arbitrary-mask model + leakage guards) and 5 training seeds, the generative reconstructor does *not* outperform a simple held-out *ridge achievability comparator* (a ridge fit on the observed leads, used as an achievability reference—*not* a Bayes-optimal oracle, since the model can and does exceed it at low guidance). The deficit $\Delta\rho = \rho_{\text{ridge}} - \rho_{\text{model}}$ —how far the model’s target-lead correlation falls below the ridge comparator ρ_{ridge} —is *monotone* in w : mean $\Delta\rho = -0.026, -0.006, +0.041, +0.072$ for $w = 1, 2, 4, 6$ (mean over 5 seeds; SD ± 0.044 at $w=6$). It is small and negative at $w=1$ (the model marginally beats the comparator) and grows to a modest but seed-*significant* $+0.072$ at $w=6$ (95% t -interval $[+0.011, +0.134]$, *excludes zero*; the model now falls short of the comparator). Guidance thus moves the gap the *wrong* way for the fabrication story: it does not buy achievability beyond a plain ridge, it erodes the model’s small low-guidance edge and drives it significantly below the comparator. The previously reported large diffusion advantage does not reproduce.

(2) Realism does not improve with guidance—the causal premise is false. The story required higher guidance to buy realism at the cost of fidelity. Instead realism is *no better* at high guidance: from $w=1$ to $w=6$ the target-lead PSD log-distance rises (V2: $1.88 \rightarrow 2.25$; a small dip at $w=2$ then a clear rise, so worsening but not monotone) and the amplitude-Wasserstein distance rises (V2: $0.45 \rightarrow 0.83$). The trend is toward *worse* realism on essentially every target-lead axis (both PSD log-distance and amplitude-Wasserstein rise from $w=1$ to $w=6$); there is no monotone realism/fidelity trade to exploit. This is consistent with fundamental limits on hallucination in inverse problems [14] and the perception–distortion tradeoff [15]: raising perceptual pressure (guidance) cannot manufacture target-lead information the observed leads do not carry.

(3) What is true instead. The predictable non-dipolar residual is real but modest: ridge-comparator correlations (gpu_oracle_gate.json) are $\rho \approx 0.49\text{--}0.55$ for V2 (QRS/ST) and $0.21\text{--}0.34$ for V4/V6—consistent with the calibrated Tier-II intervals of Sec. B. So the honest statement is a *graded predictability* of the residual, calibrated per feature and lead, not a proof of fabrication. We removed the fabrication-objective claim and rebuilt the paper around the parts that withstand scrutiny.

(4) An exploratory null-space energy diagnostic. The certificate isolates the dipole subspace $\mathbf{Q} = \mathbf{I} - \mathbf{P}_{\text{obs}}$ the observed leads cannot constrain. As a *descriptive, exploratory* diagnostic we measure the fraction of a reconstruction’s dipolar energy placed there—the *null-space dipolar energy ratio* $R_Q = \|\mathbf{M}_s \mathbf{Q} \hat{\mathbf{d}}\|^2 / \|\mathbf{M}_s \hat{\mathbf{d}}\|^2$ (Fig. 3). We are explicit about what R_Q is *not*: $R_Q > 0$ means a reconstruction asserts energy in the unidentifiable subspace, but it does *not* by itself demonstrate fabrication. The true dipole can carry $\mathbf{Q}$ -energy, and under correlated observed/unobserved dipole

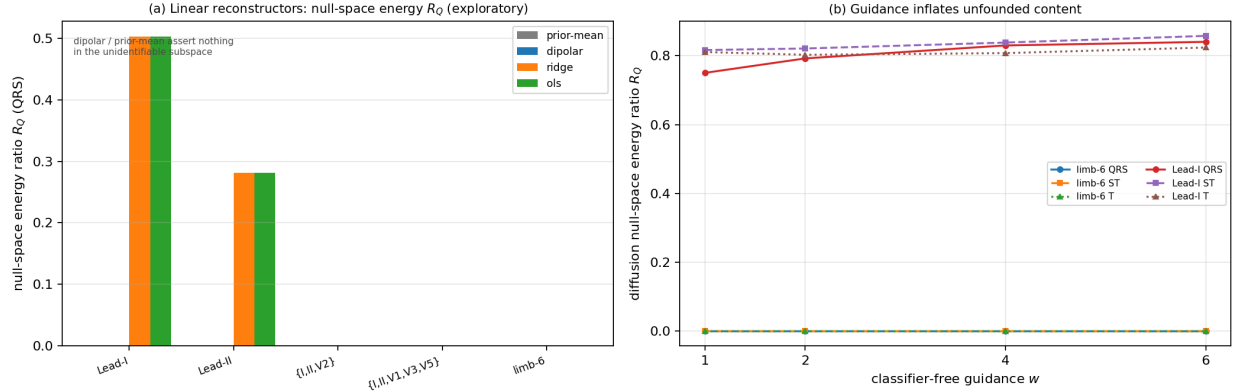


Figure 3: **Null-space dipolar energy diagnostic** (exploratory, ground-truth-free). R_Q = fraction of a reconstruction’s dipolar energy in the unidentifiable subspace $\mathbf{Q}$; $R_Q > 0$ is a descriptor, *not* proof of fabrication (Sec. H). (a) Linear reconstructors leave the unobserved coordinate near zero ($R_Q \approx 0$) except on a heavily under-determined single lead, where ridge/OLS place energy in $\mathbf{Q}$. (b) The diffusion model places most of its dipolar content in $\mathbf{Q}$ on a single lead, increasing with classifier-free guidance w ; on the rank-2 limb-6 set (which does *not* span the dipole) the fitted maps happen not to excite the one unobserved direction, so $R_Q \approx 0$. Regenerated from `results/fabrication_audit.json`, `fabrication_diffusion.json`.

coordinates the Bayes posterior mean $\mathbb{E}[\mathbf{Q}\mathbf{d} \mid \mathbf{P}\mathbf{d}]$ is generally *nonzero*; hence a reconstruction with $R_Q=0$ (which sets $\mathbf{Q}\hat{\mathbf{d}}=0$) is *not* the posterior mean and $R_Q=0$ is not “correct abstention.” A defensible metric would instead standardize the deviation $\mathbf{Q}\hat{\mathbf{d}} - \mathbb{E}[\mathbf{Q}\mathbf{d} \mid \mathbf{P}\mathbf{d}]$ by $\Sigma_{\mathbf{Q}|P}$ (Sec. A); we leave that calibrated assertion score to future work and report R_Q only as an exploratory descriptor. Descriptively: reconstructors that leave the unobserved coordinate near zero have $R_Q \approx 0$ (≤ 0.000); on a heavily under-determined single lead ridge/OLS put $R_Q = 0.50$ (Lead-I QRS) into $\mathbf{Q}$, and the diffusion model places most of its dipolar content there ($R_Q \approx 0.75$), *increasing* with guidance ($0.75 \rightarrow 0.84$ QRS, $\rightarrow 0.86$ ST as $w:1 \rightarrow 6$). The rank-2 limb-6 set does *not* span the rank-3 dipole, yet $R_Q \leq 0.00$ there for these reconstructors—an empirical observation that the fitted linear/generative maps do not excite the one unobserved limb-6 direction, *not* evidence that limb-6 is dipole-spanning. We do *not* read any of this as certified fabrication.

Ethics, data, funding, and AI-use disclosure

Data and ethics. The planned experiments use publicly available, de-identified databases distributed for research: PTB-XL [9] and the Chapman–Shaoxing–Ningbo 12-lead ECG database [10]. No new human-subject data are collected and no identifiable information is used. This methodological study is not clinical advice.

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Development-time AI-tool provenance. OpenAI Codex and Anthropic Claude Code assistants were used during repository analysis, code and manuscript review, experiment-workflow support, and prose editing. An AutoResearchClaw v0.5.0 control configuration was prepared, but the AutoResearchClaw pipeline did not produce research or review outputs because its required `acpx`

bridge was unavailable. It must not be credited as having reviewed the paper unless a later successful run is preserved with its artifacts.

Submission disclosure placeholder—complete before Stage 20. Record each tool’s provider, product, model/version, access date, affected files or tasks, prompts or run identifiers where policy requires them, and the human verification performed. Current status: PENDING PROVENANCE-LEDGER FREEZE.

AI systems are not authors. The human author remains responsible for study design, source and citation verification, code execution, statistical interpretation, every numerical value, and the submitted text. This disclosure must be reconciled with the destination venue’s policy before submission.

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